Artur Xarles i Esparraguera

Computer Vision Researcher | Sports Data Scientist

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Final-year PhD researcher combining computer vision and data science to understand sports. My work focuses on localizing and anticipating actions in video, tracking players and the ball, and assessing how actions contribute to team performance. Sports have been part of my life since childhood, both as a player and a fan. I want to combine that passion with my technical expertise to develop technologies that improve how sports are analyzed and understood.

SELECTED RESEARCH & EXPERTISE

Computer Vision

ACTION SPOTTING

Developed multiple methods to localize brief actions precisely in sports video. **AdaSpot** (CVPR 2026) introduced adaptive multi-resolution processing: low-resolution frames preserve global context while selected high-resolution regions add fine detail with little extra computation. AdaSpot achieved state-of-the-art results across several sports benchmarks.

T-DEED (CVPRW 2024) uses a temporal encoder-decoder that combines multiple time scales to capture the context needed by different action types; it also achieved state-of-the-art results across sports datasets at the time.

Also explored multimodal spotting in **ASTRA** (ACM MMSports 2023), combining video and audio, and **TraVIS** (ACM MMSports 2026), fusing video with player and ball tracking.

ACTION ANTICIPATION

Co-developed **FAANTRA** (CVPRW 2025), which introduced a football anticipation benchmark, evaluation metrics, and a transformer baseline for predicting future ball actions from past video.

TRACKING & POSE

Developed baselines and supervised MSc projects in player tracking, ball tracking, and video stitching, and explored pose estimation within the FIFA Skeletal Tracking Light Challenge.

Data Science

ACTION VALUATION

Action valuation scores player actions by their quality and contribution to a team's objectives.

Action Valuation Survey (CVPRW 2025) presented a nine-dimension taxonomy of existing methods and identified two major gaps: limited public data and a lack of standardized evaluation.

Benchmarking Action Valuation (submitted, 2026) introduced BenchBomb, a benchmark based on public data with standardized splits and an objective evaluation protocol. A systematic component analysis identified effective supervision strategies, game-state representations, and modeling architectures. These findings informed VAST, a spatio-temporal model that outperformed established action valuation methods on BenchBomb.

RESEARCH & INDUSTRY EXPERIENCE

PhD Researcher, Computer Vision & Data Science | Universitat de Barcelona — Feb 2023 – Present
Led first-author projects from model design and ablations through paper writing and public PyTorch code.

Research Stay, Computer Vision | Game On Technologies, Maastricht — Jul – Nov 2025
Developed AdaSpot; guided MSc projects on player/ball tracking and multi-camera video stitching using professional football footage.

Software Engineer, Computer Vision | Computer Vision Center — Sep 2022 – Jun 2023
Developed sports-video action spotting models using temporal, multimodal, and self-supervised learning.

EDUCATION

PhD in Computer Science | Universitat de Barcelona — 2023 – expected 2027
Thesis submission planned Dec 2026.

MSc in Fundamental Principles of Data Science | Universitat de Barcelona — 2021 – 2022

BSc in Applied Statistics | Universitat Autònoma de Barcelona — 2017 – 2021

LEADERSHIP, TEACHING & SERVICE

- Coordinated **4 master's theses**; visiting lecturer in the Master in Computer Vision, Universitat de Barcelona.
- Invited speaker, **FIFA Research Institute at KAUST**; co-organizer, **12th CVsports Workshop at CVPR 2026**.
- **SoccerNet task organizer:** 2025 Team Ball Action Spotting and 2026 Ball Action Anticipation challenges.

TECHNICAL SKILLS & LANGUAGES

Programming & tools: Python, R, SQL, PyTorch, Git, Linux. | **Languages:** Catalan, Spanish, English.
Methods & data: deep learning, spatio-temporal modeling, multimodal video, event and tracking data.